Software Requirements Specification for Software Engineering: subtitle describing software

Team 11, technically functional
Matthew Huynh
Cieran Diebolt
Vaisnavi Shanthamoorthy
Maham Siddiqui
Eman Ashraf

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Contents

Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

Introduction

The software requirements specification (SRS) document describes the functional and non-functional requirements, scope and functionality of the application. It formally outlines key measures relevant to all the different stages of the software development lifecycle. This document is constructed using the Volere Software Requirements Specification template [?]

1 Purpose of the Project

1.1 User Business

Insert your content here.

1.2 Goals of the Project

Insert your content here.

2 Naming Conventions and Terminology

Insert your content here.

3 Stakeholders

The stakeholders for this project are parties with a vested interest in the end product. They will be consulted on their requirements and opinions throughout the development process. Their impact will be foundational for correctness of the product. This section outlines the various types and levels of stakeholders, the personas, user participation (towards application development) and any users that will be responsible for the product's maintenance.

3.1 Client

There are two categories of clients that are the target audience for this product. The first category of end users is the patients. The application is being designed for patients who are undergoing physiotherapy treatment and have been given a home exercise program to aid in their rehabilitation. For the purpose of this system, the exercise and the joint selected was recommended by one of our stakeholders. The knee and a straight leg raise were chosen.

The second category of the clients are the physiotherapists who are treating the users on the application. It will integrate the physiotherapist's patient list and their corresponding performance statistics. The application allows physiotherapists to gain insight into how their patients' are progressing. Furthermore, it will enable the assessment of the suitability of the prescribed exercise, as well as gauging whether any additional modifications are required (number of repetitions, degree of complexity etc.).

3.2 Customer

The application is being developed for a Canadian market with its governing standards and regulations in mind. Although this version of the application does not have any customers, it can prospectively be purchased by any healthcare provider and be incorporated into their patient care structure.

3.3 Other Stakeholders

The primary and secondary stakeholders are described above (clients and customers).

Other stakeholders may include regulatory authorities or other healthcare specialists, such as physiatrists or massage therapists. The regulatory authorities will be allowed access to conduct audits and ensure that any disclosed patient information is being handled in an ethical manner. In addition, they will verify the accuracy and quality of care being delivered. Other healthcare providers may benefit from any insights or information that is being provided, which may be used to tailor their own care of the patient. Lastly, support workers may use this application to assist their clients with their home exercises.

3.4 Hands-On Users of the Project

<u>**Patients**</u>
User role: Record corresponding exercise for application to analyze, adjust exercise form based on application feedback and correction, gain insight from performance metrics provided by the application.
Subject matter experience: Aware of how to perform exercise using the

instructions provided by the physiotherapist (eg. a worksheet detailing the steps required for the exercise).
Technological experience: Know how to operate the smart device and use its camera to record their performance of the exercise.
Attitude towards technology: Open to learning if needed.
Physical abilities/disabilities: Able to use a smart device and/or perform the prescribed exercise.
Physiotherapists
User role: Access patient statistics to assess suitability of exercise, difficulty and whether the prescribed exercise is effective.
Subject matter experience: Formal education on the topic and have knowledge of general and post operative care.
Technological experience: Able to effectively use the various functionalities of the application.
Attitude towards technology: Varied.
Education: Officially licensed.

3.5 Personas

Name: Scotty Fye
Age: 72
Occupation: Retired (Former teacher)
Location: Toronto, Canada
Status: Married, with children

"I had knee surgery a few weeks ago because my knee pain was unbearable. As suggested by my doctor, I had knee replacement surgery. Now, I am following some prescribed exercises but I'm worried that I am doing them wrong and possibly slowing down recovery. On top of that, I have a vacation with my family and want to be the 'fun grandpa'".

- Goals:
    - To gain mobility and strength in his knee for an upcoming vacation and keep up with his active grandchildren.

- He wants to understand how to both manage pain and recover well.
  - See how he is performing on a weekly basis.
  - Have guidance for performing these exercises alone to ensure he is performing exercises correctly.

- Fear:
  - Worried about doing something wrong which might cause pain and hinder his recovery.
  - Prescribed exercises are manageable but is unsure if he should push through discomfort.
  - "Should I push through stiffness or should I stop?"
  - "Am I bending my knee enough? Not enough?"

Name: Maya Thompson
Age: 21
Occupation: Student
Location: Hamilton, Ontario

"I was in a car accident which shattered my knee. I had to get a full knee replacement in order to continue walking. I was recently discharged from a rehabilitation facility where I started learning how to use my knee and walk again. My physiotherapist enrolled me as their patient on this application to keep up to date on my progress towards recovery."

- Goal:
  - To gain sufficient use of her knee to walk and perform activities of daily living.
  - To monitor her own progress and use it as a motivating factor for pushing herself.
  - To gain enough strength in her muscles without having to further exacerbate her pain.

- Fear:
  - Her muscles need to relearn how to function and she is afraid if she performs an exercise incorrectly, the incorrect muscles will learn to compensate during the movement.
  - ‘Am I using the correct muscles? Is this the muscle that I need to consciously activate?’
  - Her pain might be aggravated to an intolerable degree.
  - ‘What if my pain does not start out as much but becomes unbearable because I did not perform the exercise correctly?’

3.6 Priorities Assigned to Users

Key users: Patients undergoing post operative physiotherapy have highest priority in the system, physiotherapists using the system to check-in and comment on patient exercises have similar priority.

3.7 User Participation

Users (Patients)

- User creates an account and set up their profile and may sign into this account with valid credentials.
- User requests analytical information based on their past exercise inputs and results.
- User receives live form recommendations from the system by providing a video-feed of themselves performing the selected activity.
- User requests an overview of their exercise after successfully completing a video-feed upload, or looks back at old overviews.
- User creates comments on their own exercise overviews to keep personal notes and information regarding exercise session.
- User receives comments from the physiotherapist account linked to theirs.

Users (Physiotherapists)

- User creates an account and sets up their profile, and may sign into this account with valid credentials.
- User requests a list of all patient accounts connected to their account.
- User receives overviews of patient's exercises to confirm exercise completion and progress on recovery.
- User creates comments on patient's exercise overviews to assist the patient with exercise performance, and/or to provide modifications.

3.8 Maintenance Users and Service Technicians

Maintenance and service on the system will be performed by the current development team. Maintenance and service should be minimized with a thorough test suite to ensure all functionalities of the system are operational upon any changes or updates.

4 Mandated Constraints

4.1 Solution Constraints

Insert your content here.

4.2 Implementation Environment of the Current System

Insert your content here.

4.3 Partner or Collaborative Applications

Insert your content here.

4.4 Off-the-Shelf Software

Insert your content here.

4.5 Anticipated Workplace Environment

Insert your content here.

4.6 Schedule Constraints

Insert your content here.

4.7 Budget Constraints

Insert your content here.

4.8 Enterprise Constraints

Insert your content here.

5 Naming Conventions and Terminology

5.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

Insert your content here.

6 Relevant Facts And Assumptions

6.1 Relevant Facts

Insert your content here.

6.2 Business Rules

Insert your content here.

6.3 Assumptions

Insert your content here.

7 The Scope of the Work

7.1 The Current Situation

Insert your content here.

7.2 The Context of the Work

Insert your content here.

7.3 Work Partitioning

Insert your content here.

7.4 Specifying a Business Use Case (BUC)

Insert your content here.

8 Business Data Model and Data Dictionary

8.1 Business Data Model

Insert your content here.

8.2 Data Dictionary

Insert your content here.

9 The Scope of the Product

9.1 Product Boundary

Insert your content here.

9.2 Product Use Case Table

Insert your content here.

9.3 Individual Product Use Cases (PUC's)

Insert your content here.

10 Functional Requirements

10.1 Functional Requirements

Insert your content here.

11 Look and Feel Requirements

11.1 Appearance Requirements

Insert your content here.

11.2 Style Requirements

Insert your content here.

12 Usability and Humanity Requirements

12.1 Ease of Use Requirements

Insert your content here.

12.2 Personalization and Internationalization Requirements

Insert your content here.

12.3 Learning Requirements

Insert your content here.

12.4 Understandability and Politeness Requirements

Insert your content here.

12.5 Accessibility Requirements

Insert your content here.

13 Performance Requirements

13.1 Speed and Latency Requirements

Insert your content here.

13.2 Safety-Critical Requirements

Insert your content here.

13.3 Precision or Accuracy Requirements

Insert your content here.

13.4 Robustness or Fault-Tolerance Requirements

Insert your content here.

13.5 Capacity Requirements

Insert your content here.

13.6 Scalability or Extensibility Requirements

Insert your content here.

13.7 Longevity Requirements

Insert your content here.

14 Operational and Environmental Requirements

14.1 Expected Physical Environment

Insert your content here.

14.2 Wider Environment Requirements

Insert your content here.

14.3 Requirements for Interfacing with Adjacent Systems

Insert your content here.

14.4 Productization Requirements

Insert your content here.

14.5 Release Requirements

Insert your content here.

15 Maintainability and Support Requirements

15.1 Maintenance Requirements

Insert your content here.

15.2 Supportability Requirements

Insert your content here.

15.3 Adaptability Requirements

Insert your content here.

16 Security Requirements

16.1 Access Requirements

Insert your content here.

16.2 Integrity Requirements

Insert your content here.

16.3 Privacy Requirements

Insert your content here.

16.4 Audit Requirements

Insert your content here.

16.5 Immunity Requirements

Insert your content here.

17 Cultural Requirements

17.1 Cultural Requirements

Insert your content here.

18 Compliance Requirements

18.1 Legal Requirements

Insert your content here.

18.2 Standards Compliance Requirements

Insert your content here.

19 Open Issues

Insert your content here.

20 Off-the-Shelf Solutions

20.1 Ready-Made Products

Insert your content here.

20.2 Reusable Components

Insert your content here.

20.3 Products That Can Be Copied

Insert your content here.

21 New Problems

21.1 Effects on the Current Environment

Insert your content here.

21.2 Effects on the Installed Systems

Insert your content here.

21.3 Potential User Problems

Insert your content here.

21.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Insert your content here.

21.5 Follow-Up Problems

Insert your content here.

22 Tasks

22.1 Project Planning

Insert your content here.

22.2 Planning of the Development Phases

Insert your content here.

23 Migration to the New Product

23.1 Requirements for Migration to the New Product

Insert your content here.

23.2 Data That Has to be Modified or Translated for the New System

Insert your content here.

24 Costs

Currently, the cost of this project is 0 dollars. However, the only cost at this moment is the team members' time and expertise in order to bring this project to fruition. All the tools that are currently being used to build the product are free to use and open source.

24.1 Metrics for Estimation

Financial costs are not a concern at this stage. However, circumstances may change and require re-evaluation of any cost metrics to include this new development. In that case, a cost-benefit analysis would be performed in

order to determine the correct value.
The metrics listed below estimate the time and effort put forth by a team member:

- The *quality* of commits and pushes - this can be assessed by the commit message's clarity and descriptiveness.
- The *quantity* of contributions made to the project is an additional indicator of effort. However, it should be contextualized in terms of quality.
- The quality of other deliverables can be similarly analyzed.

As a team, the collective time and effort will be measured by:

- The number of goals that have been achieved.
- The amount of stretch goals that may have been successfully achieved.
- The quality of the final project deliverables, measured using deliverable grades and TA's feedback.
- Number of test cases that have been devised.
- Cohesiveness of the final product, which can be quantified by how functional the final application is.
- The amount of time spent on a project deliverable.

24.2 Estimation Approach

Current estimations are based on the previous and ongoing deliverable and may be inaccurate. They are subject to modification throughout the project timeline.

24.3 Cost Breakdown

Tools and Software:
The cost is 0 dollars since all libraries and additional software will be open source.
Testing Environment:

The projected test suite will be formulated in house and will have no cost.
Development effort:

- Documentation (generating output, editing): 120 hours per member
- Deliverable revision: 20 hours per member
- Team meetings: 50 hours per member
- Software development: 400 hours per member
- Software testing: 100 hours per member

Total: 690 hours per member

25 User Documentation and Training

25.1 User Documentation Requirements

Insert your content here.

25.2 Training Requirements

Insert your content here.

26 Waiting Room

Insert your content here.

27 Ideas for Solution

Insert your content here.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?